



AIMS

African Institute for
Mathematical Sciences
SOUTH AFRICA

PEDAGOGICAL CONCEPTS

Nigeria - Aug 2026

AI Research Foundations

with support from

Google.org

What Level Were You Thinking At?

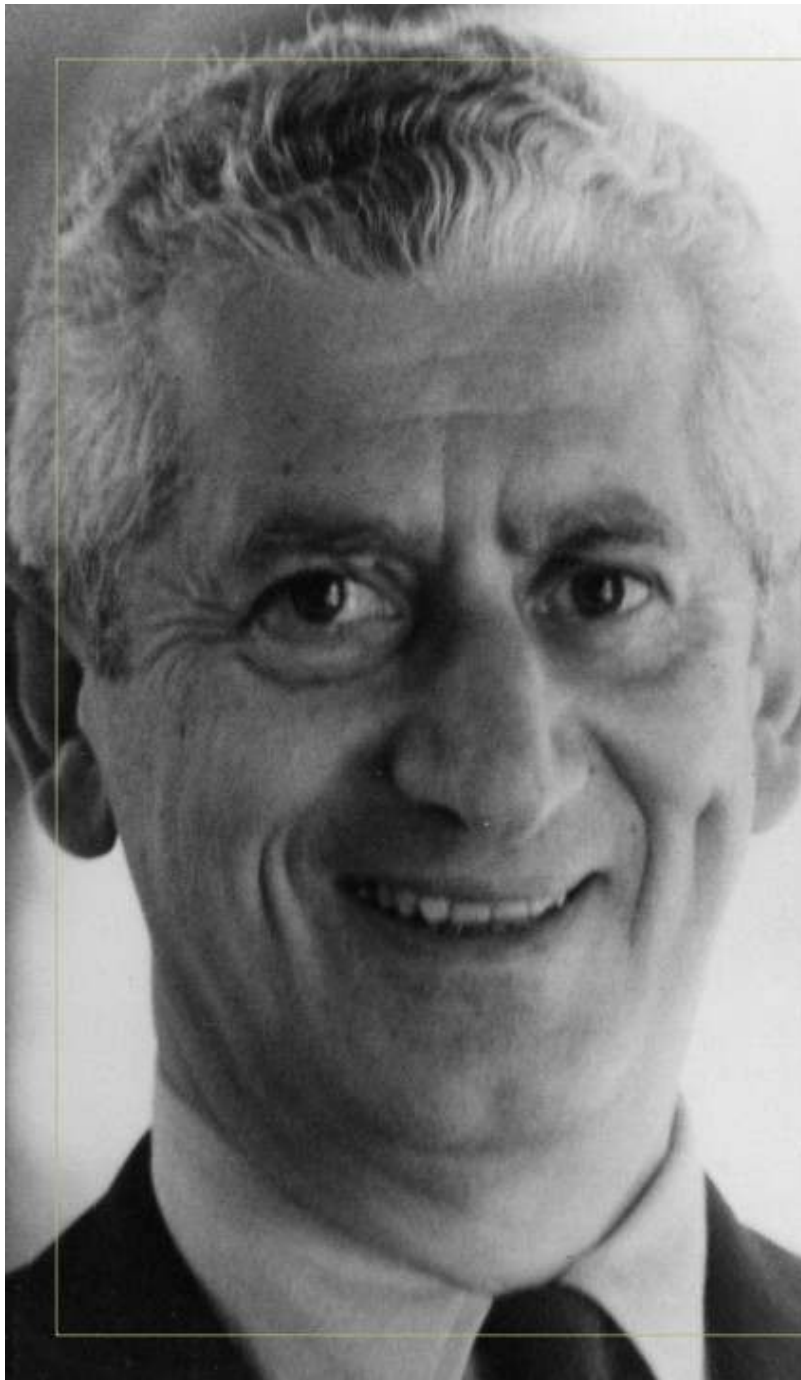


'Jide was hungry so she went looking for...'

...and predicted the next word. What were you doing, cognitively?

Recalling a fact? Applying a rule? Creating something new?

Decorative yellow concentric circles in the bottom right corner of the slide.



Benjamin BLOOM



EDUCATIONAL
PSYCHOLOGIST

edutopia



Learning Objective

DEFINITION A clear, specific statement describing what a learner is expected to be able to do by the end of a lesson, course, or activity : always starting with an action verb.

Examples:

- By the end of this lesson, students will be able to understand tokenization and compare different tokenization methods.
- By the end of this course, students will be able to explain how the BPE algorithm builds a vocabulary step by step.

Three Domains of Bloom's Taxonomy

COGNITIVE

How learners
process knowledge

IN SCOPE TODAY

AFFECTIVE

How learners engage
with values, attitudes
and ethics

IN SCOPE TODAY

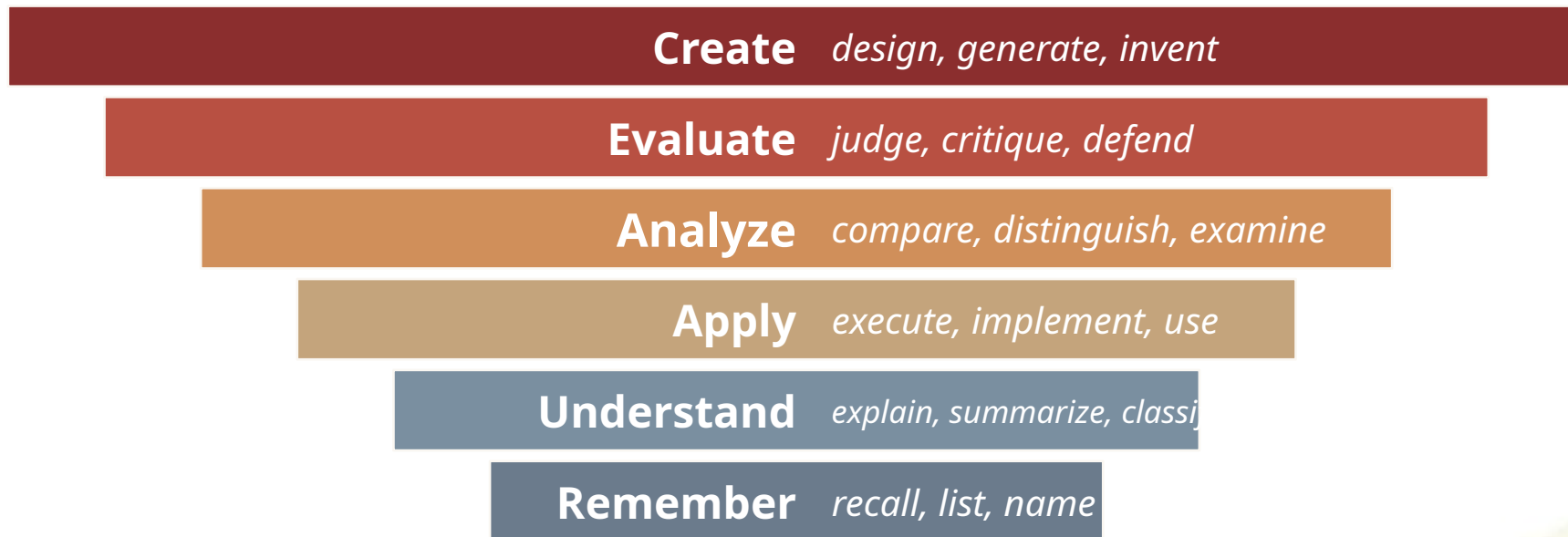
PSYCHOMOTOR

How learners develop
physical skills through
practice and hands-on
experience

out of scope

The Cognitive Domain

DEFINITION Describes how learners develop knowledge and skills, from remembering facts to creating original work.



Cognitive Domain

REMEMBER · L1

The ability to remember and recall previously learned facts, concepts, or skills.

Example

List the five steps of the BPE algorithm used to create subword tokens from text data.

List

Recall

Name

UNDERSTAND · L2

The ability to explain information in your own words, demonstrating understanding of its meaning.

Example

Explain why word-level tokenization struggles with unknown words.

Explain

Summarize

Describe

Cognitive Domain

APPLY · L3

The ability to apply knowledge in new situations by using what you know to solve problems.

Example

Apply the first three steps of the BPE algorithm to a small corpus.

Implement

Use

Solve

ANALYZE · L4

The ability to break complex information into parts and examine how those parts connect and relate to each other.

Example

Distinguish between the OOV problem in word-level tokenization and the long sequence problem in character-level tokenization.

Compare

Distinguish

Examine

Cognitive Domain

EVALUATE · L5

The ability to make informed judgments using criteria and standards to assess the quality or validity of something.

Example

Evaluate whether removing emojis from review data is a good preprocessing choice for a sentiment analysis task.

Judge

Assess

Argue

CREATE · L6

The highest level of thinking, involving combining elements to create something new, such as a product, plan, or idea.

Example

Design a 2D embedding map for ten words from your own language community.

Design

Build

Plan

The Affective Domain

DEFINITION The affective domain is about how learners develop values and ethics, from receiving information to internalizing it as part of their behaviour.

Higher-order →

Internalize *embody, act consistently*

Organize *integrate, prioritize*

Value *advocate, commit*

Respond *react, comply*

Receive *attend, observe*

Affective Domain

RECEIVE · L1

Students become aware of an attitude, behaviour, or value and show willingness to pay attention to it.

Example

Listen to a community member describe how they use an AI tool in daily life.

Attend

Observe

Notice

RESPOND · L2

Students respond to a value or attitude and demonstrate active participation.

Example

Discuss how a community is affected when an AI model does not understand their language.

Discuss

Contribute

Share

Affective Domain

VALUE · L3

Students recognise and demonstrate a value through their involvement or commitment, showing that they care about it.

Example

Advocate for community members having a say in how their language and culture are represented in AI.

Advocate

Commit

Justify

ORGANIZE · L4

Students determine priorities and resolve conflicts between different values, developing a personal value system.

Example

Prioritize local languages over convenience when choosing training data for an African AI project.

Prioritize

Integrate

Balance

Affective Domain

INTERNALIZE · L5

Students act consistently with their internalized beliefs, making the value a natural part of their behaviour.

Example

Embody the habit of asking who is affected by your AI tool before asking how well it performs.

Embody

Maintain

Influence

Online vs. Classroom Learning Objectives

The learning objective of online learning is to build basic knowledge, while the learning objective of classroom learning is to develop deeper thinking and more complex skills.

ONLINE COURSES

Remember (L1)

List the types of tokenization: sentence-level, word-level, character-level, and subword tokenization.

CLASSROOM SESSIONS

Analyze (L4)

Compare how two tokenizers split a word and identify which one better preserves its meaning.

AI vs RI Activities

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Two domains, two ladders

COGNITIVE

Remember	Retrieve
Understand	Explain
Apply	Execute a procedure
Analyze	Tune hyperparameters; compare training curves
Evaluate	Judge competing design choices
Create	Design an MLP architecture

AFFECTIVE

Receive	Listen / attend
Respond	React / participate
Value	Demonstrate commitment to something
Organize	Integrate different values
Internalize	Integrate values into professional practice

The UCL Lecturer Toolkit

01 (GCSB)

Before class - Online

Students complete the Spotlight activities on Google Cloud Skills Boost. Each one is tagged with a Conversational Framework verb and a Bloom level.

02 (Extend - UCL)

In class - Activity menu

The lecturer builds the session from the toolkit's pool of AI (cognitive) and RI (affective) activities : all on UCL Extend.

The Toolkit on UCL Extend

UCL Extend · extend.ucl.ac.uk

- Session 0 ... Session 9 (one course page per session)
 - Session guide – handbook-style wrap-around material
 - AI activities · Short / Medium / Long
 - RI activities · Short / Medium
 - Slides – ready-to-present lecturer deck
 - Notebooks – Student + Teacher (.ipynb)

One home for everything : UCL Extend



Slides

10 decks

One ready-to-present lecturer deck for every session, 0 through 9.



Learning Activities

10 session pages

AI and Responsible Innovation activities for each session, in Short, Medium and Long formats, wrapped in handbook-style guidance.

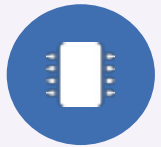


Notebooks

21 files

Jupyter labs for the AI Long activities : in Student and Teacher versions plus real datasets.

Toolkit activities: AI & RI



AI Technical skills

- Machine-learning foundations
- Embeddings and neural networks
- Mini-hackathons and end-to-end labs
- Formats: Short - Medium - Long
- Long activities come with lab notebooks



RI Responsible innovation

- Ethics and responsibility in AI
- Societal impact and fairness
- Reflection and structured discussion
- Formats: Short - Medium
- Facilitation guides

AI Long labs & notebooks

AI Long activity

Lab notebook · Student + Teacher

1.7 Machine-learning lab



1.7 Lab notebook · **Student + Teacher** .ipynb

2.6 Mini-hackathon



2.6 Lab notebook · **Student + Teacher** .ipynb

2.7A Small-embeddings lab



2.7A Lab notebook · **Student + Teacher** .ipynb

3.7 Neural Network Training



3.7 Lab notebook · **Student + Teacher** .ipynb

4.6A From-embeddings lab



4.6A Lab notebook · **Student + Teacher** .ipynb

The Two Pillars

AI Activities

Cognitive domain - Technical - Hands-on

“How does it work?”

Asks learners to implement, compute, train, compare, or design something technical.

RI Activities

Affective domain - Ethical - Reflective

“Who does it affect - and why does that matter?”

Asks learners to reflect, discuss, evaluate values, map stakeholders, or surface missing voices.



Course 1 - Build Your Own SLM

AI Activities

Experiment with N-grams

Build a probabilistic next-word predictor and test its limits on a small corpus.

RI Activities

Applying a local ethical lens to the trolley problem

Explore your own ethical framework: thoughts, feelings, and values.

Course 2 - Represent Your Language Data

AI Activities

Lab: Implement a BPE Tokenizer

Iteratively merge subword tokens on a small corpus to see how modern tokenizers learn.

RI Activities

Whose voice is missing?

Reflect on the exclusion introduced by dataset selection when training LLMs.

Course 3 - Design & Train Neural Networks

AI Activities

Lab: Mitigate Overfitting

Apply early stopping, dropout and regularization to build models that generalise.

RI Activities

Anticipating risks

What can be learned from the challenges identified for LLMs in the research literature?

Course 4 - Discover the Transformer Architecture

AI Activities

Lab: Implement Masked Multi-Head Attention

Extend a single attention head and investigate which dependencies different heads capture.

RI Activities

Community values and meaning in an automated world

How do human interactions create shared meaning, and what may be lost when automated?

Course 5 - Fine-Tune Your Model

AI Activities

How will your model respond?

Work out by hand how to format prompts for different tasks.

RI Activities

Imagining AI in cultural contexts

Write a short story exploring local values and meanings around AI in your context.

Course 7 - Accelerate Your Model

AI Activities

Smart numbers

Reason through how lower-precision numbers let you train bigger, more powerful models.

RI Activities

Environmental impacts of LLMs

Map how stakeholders are affected by the environmental costs of fine-tuning LLMs.

Classify an Activity, AI? RI? Which level?

Example 1

“Train a convolutional neural network to classify maize leaf images as healthy, blighted, or rust-infected using a labelled dataset of 5,000 photographs from Kenyan smallholder farms”

AI

Apply

Example 2

“If your model misdiagnoses a smallholder farmer's only crop, who bears the cost?”

RI

Respond

AI Activity or RI Activity?
Which level in Bloom's Taxonomy?



Question 1 of 10 · AI or RI? Which level?

“Implement a BPE tokenizer from scratch and test it on isiZulu text.”

A. AI Activity / Apply

B. RI Activity / Apply

C. AI Activity / Respond

D. AI Activity / Create



Question 1 • Answer

“Implement a BPE tokenizer from scratch and test it on isiZulu text.”

A. ✓ AI Activity / Apply

B. RI Activity / Apply

C. AI Activity / Respond

D. AI Activity / Create

Question 2 of 10 · AI or RI? Which level?

“Discuss who owns the data when companies scrape African language websites. Use the Fair Trade analogy to propose a governance model.”

A. AI Activity / Value

B. RI Activity / Respond

C. RI Activity / Value

D. AI Activity / Analyze



Question 2 • Answer

“Discuss who owns the data when companies scrape African language websites. Use the Fair Trade analogy to propose a governance model.”

A. AI Activity / Value

B. ✓ RI Activity / Respond

C. RI Activity / Value

D. AI Activity / Analyze



Question 3 of 10 · AI or RI? Which level?

“Compare models trained for 10, 400, and 1000 epochs. Identify overfitting and underfitting in the loss curves.”

A. AI Activity / Apply

B. AI Activity / Analyze

C. RI Activity / Value

D. AI Activity / Value



Question 3 • Answer

“Compare models trained for 10, 400, and 1000 epochs. Identify overfitting and underfitting in the loss curves.”

A. AI Activity / Apply

B. ✓ AI Activity / Analyze

C. RI Activity / Value

D. AI Activity / Value



Question 4 of 10 · AI or RI? Which level?

“Watch the Masakhane video on low-resource languages. Write a paragraph on what 'harnessing potential' means for your institution.”

A. AI Activity / Understand

B. RI Activity / Respond

C. AI Activity / Respond

D. RI Activity / Value



Question 4 • Answer

“Watch the Masakhane video on low-resource languages. Write a paragraph on what 'harnessing potential' means for your institution.”

A. AI Activity / Understand

B. RI Activity / Respond

C. AI Activity / Respond

D. ✓ RI Activity / Value

Question 5 of 10 · AI or RI? Which level?

“Estimate the FLOPs needed to fine-tune an SLM. Experiment with bfloat16 to reduce compute cost.”

A. AI Activity / Apply

B. RI Activity / Respond

C. AI Activity / Understand

D. AI Activity / Organize



Question 5 • Answer

“Estimate the FLOPs needed to fine-tune an SLM. Experiment with bfloat16 to reduce compute cost.”

A. ✓ AI Activity / Apply

B. RI Activity / Respond

C. AI Activity / Understand

D. AI Activity / Organize



Question 6 of 10 · AI or RI? Which level?

“Read Adebara et al. (2025). Identify three communities missing from a popular NLP benchmark and propose how to include them.”

A. AI Activity / Analyze

B. RI Activity / Respond

C. RI Activity / Organize

D. AI Activity / Organize



Question 6 • Answer

“Read Adebara et al. (2025). Identify three communities missing from a popular NLP benchmark and propose how to include them.”

A. AI Activity / Analyze

B. RI Activity / Respond

C. ✓ **RI Activity / Organize**

D. AI Activity / Organize



Question 7 of 10 · AI or RI? Which level?

“Implement the LoRA adapter and fine-tune an SLM on a local language dataset.”

A. AI Activity / Understand

B. AI Activity / Apply

C. RI Activity / Apply

D. AI Activity / Value



Question 7 • Answer

“Implement the LoRA adapter and fine-tune an SLM on a local language dataset.”

A. AI Activity / Understand

B. ✓ AI Activity / Apply

C. RI Activity / Apply

D. AI Activity / Value



Question 8 of 10 · AI or RI? Which level?

“A Lagos hospital wants to deploy an AI diagnostic tool trained for 1000 epochs with no validation set. Role-play a meeting between the developer, a nurse, and a patient advocate.”

A. AI Activity / Analyze

B. AI Activity / Value

C. RI Activity / Respond

D. RI Activity / Value



Question 8 • Answer

“A Lagos hospital wants to deploy an AI diagnostic tool trained for 1000 epochs with no validation set. Role-play a meeting between the developer, a nurse, and a patient advocate.”

A. AI Activity / Analyze

B. AI Activity / Value

C. RI Activity / Respond

D. ✓ RI Activity / Value



Question 9 of 10 · AI or RI? Which level?

“Design an MLP architecture for sentiment analysis on Twi product reviews. Justify your hidden layers and activation functions.”

A. AI Activity / Apply

B. AI Activity / Create

C. RI Activity / Analyze

D. AI Activity / Organize



Question 9 • Answer

“Design an MLP architecture for sentiment analysis on Twi product reviews. Justify your hidden layers and activation functions.”

A. AI Activity / Apply

B. ✓ AI Activity / Create

C. RI Activity / Analyze

D. AI Activity / Organize



Question 10 of 10 · AI or RI? Which level?

“Map the stakeholders affected by a gendered chatbot deployed by a Nigerian bank. Design a two-page community engagement plan.”

A. AI Activity / Create

B. RI Activity / Respond

C. RI Activity / Value

D. RI Activity / Organize



Question 10 • Answer

“Map the stakeholders affected by a gendered chatbot deployed by a Nigerian bank. Design a two-page community engagement plan.”

A. AI Activity / Create

B. RI Activity / Respond

C. RI Activity / Value

D. ✓ RI Activity / Organize

Three takeaways

01

Two domains, both essential.

AI activities build technical skill in the cognitive domain. RI activities build ethical reasoning in the affective domain.

02

Longer activities, higher levels.

Bloom has two relevant domains: cognitive for AI, affective for RI. Longer activities target higher levels in both.

03

Protect the pairing.

Every toolkit session pairs AI and RI intentionally. As facilitators, protect that pairing in your own course delivery.